

# Canaries in the Coal Mine? Six Facts about the Recent Employment Effects of Artificial Intelligence

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## Abstract

This paper examines changes in the labor market for occupations exposed to generative artificial intelligence using high-frequency administrative data from the largest payroll software provider in the United States. We present six facts that characterize these shifts. We find that since the widespread adoption of generative AI, early-career workers (ages 22-25) in the most AI-exposed occupations have experienced a 13 percent relative decline in employment even after controlling for firm-level shocks. In contrast, employment for workers in less exposed fields and more experienced workers in the same occupations has remained stable or continued to grow. We also find that adjustments occur primarily through employment rather than compensation. Furthermore, employment declines are concentrated in occupations where AI is more likely to *automate*, rather than *augment*, human labor. Our results are robust to alternative explanations, such as excluding technology-related firms and excluding occupations amenable to remote work. These six facts provide early, large-scale evidence consistent with the hypothesis that the AI revolution is beginning to have a significant and disproportionate impact on entry-level workers in the American labor market.

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¶Latest version: <https://digitaleconomy.stanford.edu/publications/canaries-in-the-coal-mine/>

Quarterly Census of Employment and Wages (QCEW).<sup>10</sup> These prior papers use data that lack either sufficient granularity or immediacy to reliably study employment changes by AI exposure and age (O’Brien, 2025).<sup>11</sup> In contrast, this paper uses large-scale, close to real-time data to take a step towards resolving the ongoing debate on the employment effects of AI on young workers.

## 3 Data Description

### 3.1 Payroll Data

This study uses data from ADP, the largest payroll processing firm in America. The company provides payroll services for firms employing over 25 million workers in the US. We use this information to track employment changes for workers in occupations measured as more or less exposed to artificial intelligence.

We make several sample restrictions for our main analysis sample. We include only workers employed by firms that use ADP’s payroll product to maintain worker earnings records. We also exclude employees classified by firms as part-time from the analysis and subset to people between the age of 18 and 70.<sup>12</sup>

The set of firms using payroll services changes over time as companies join or leave ADP’s platform. We maintain a consistent set of firms across our main sample period by keeping only companies that have employee earnings records for each month from January 2021 through July 2025.

In addition, ADP observes job titles for about 70% of workers in its system. We exclude workers who do not have a recorded job title. There are over 7,000 standardized job titles, examples of which include “Search engineer optimization specialist,” “Enterprise content management manager,” and “Plant documentation control specialist.” The company’s internal research team maps each of

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<sup>10</sup>Johnston and Makridis (2025) measure state-industry exposure by taking an average of Eloundou et al. (2024)’s occupational exposure weighted by state-industry employment. Industry-level labor market changes may be distinct from the occupation-level changes studied in this paper if firms make capital investments or become more productive in ways that increases overall labor demand (Hampole et al., 2025).

<sup>11</sup>As a comparison, the CPS surveyed between 44,000 and 51,000 employed individuals in total across all age groups in each month since 2021. Between 10,000 and 12,000 of these observations were in the outgoing rotation group and included earnings records. The data in our main analysis sample includes between 250,000 and 350,000 employed individuals in each month *just between the ages of 22 and 25*, all with earnings records.

<sup>12</sup>While we observe the year of birth for each worker, for privacy reasons we do not observe the exact date of birth. We impute month of birth from the distribution of birth months in the United States using data from the Center for Disease Control and Prevention.

Figure 3 shows these patterns hold more generally across professions. The top left plot shows a divergence in employment outcomes for more and less exposed occupations for workers aged 22-25, with more exposed occupations experiencing declining employment. For older age groups, we find much less marked differences in employment growth across AI exposure quintiles.

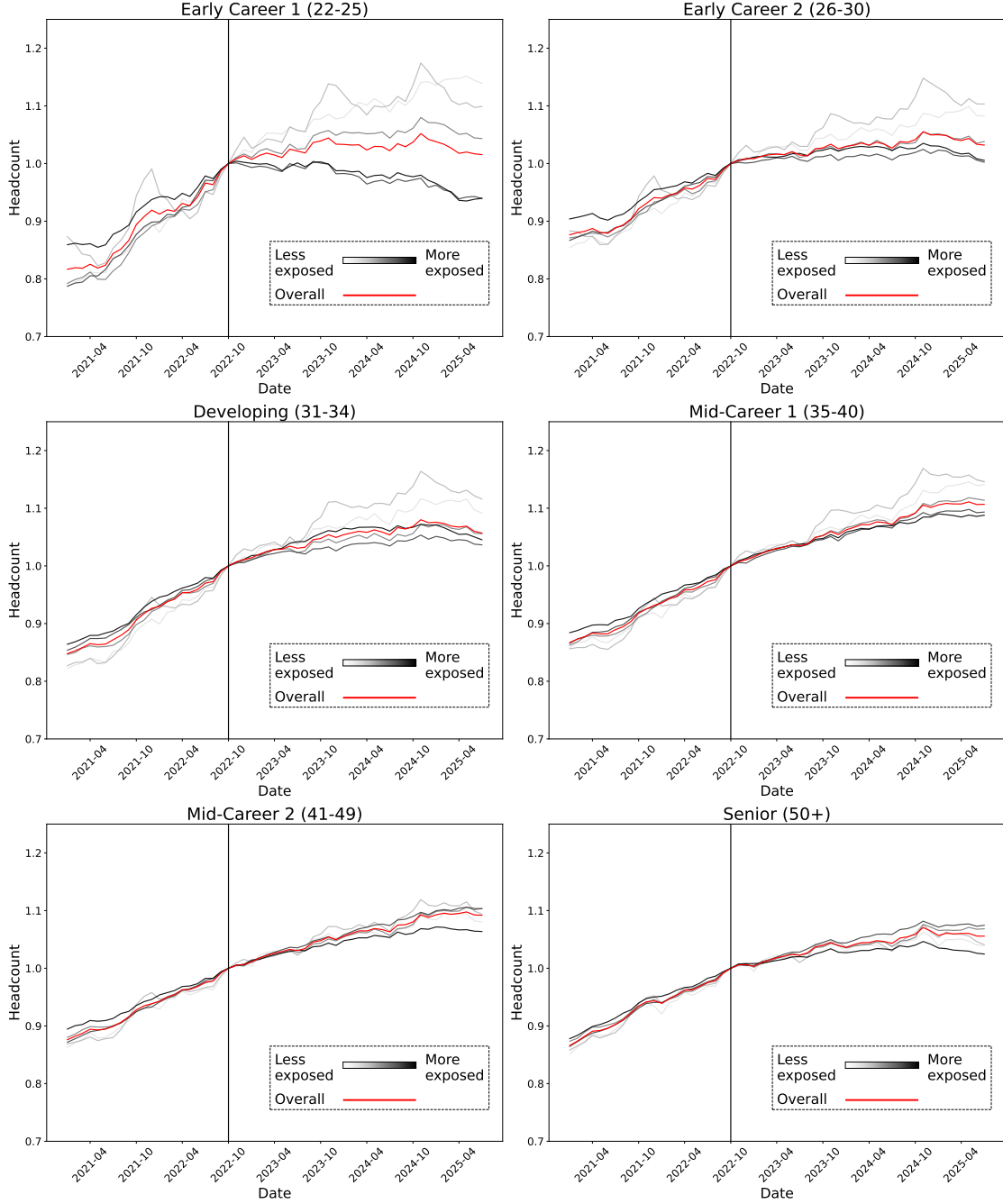


Figure 3:  $GPT-4 \beta$ . Employment changes by age and exposure quintile using measures from Eloundou et al. (2024). Exposure quintiles are defined based on the  $GPT-4 \beta$  measures. Darker lines are more exposed quintiles. The red line shows the overall trend pooling across quintiles.

$f$  indexes firms,  $q$  indexes Eloundou et al. (2024) exposure quintiles, and  $t$  indexes months, with  $t = -1$  corresponding to October 2022. The outcome variable  $y_{f,q,t}$  is employment in  $f, q, t$ . Equation 4.1 is a Poisson event study regression controlling for firm-quintile effects,  $\alpha_{f,q}$ , and firm-time effects,  $\beta_{f,t}$ . The firm-time effects absorb aggregate firm shocks that impact each exposure quintile equally. The firm-quintile effects adjust for baseline differences in hiring across quintiles within the firm. The coefficients of interest,  $\gamma_{q,t}$ , measure differential changes in employment growth across quintiles after accounting for firm-time effects and firm-quintile effects.<sup>17</sup>

We run this regression separately for each age group. For each regression, we restrict to firms that hire at least 10 workers within the age group in every period of the sample. Further,  $\sum_t y_{f,q,t}$  must equal at least 100 for each  $q$ , meaning that the firm must at least employ on average about 2 workers from each exposure quintile across months in the sample.<sup>18</sup> Standard errors are clustered by firm.

Results are in Figure 9, which plots the  $\gamma_{q,t}$  coefficients for each age group. For workers aged 22-25, estimates for higher quintiles are large and statistically significant, with a 12 log point decline in relative employment comparable in magnitude to the estimates in the raw data in Figure 3. Estimates for other age groups are generally much smaller in magnitude and not statistically significant. These findings imply that the employment trends we observe are not driven by differential shocks to firms that employ a disproportionate share of AI-exposed young workers.

One alternative confounder that would not be controlled for with firm-time effects is that even conditional on the firm workers with high AI exposure were excessively hired after the COVID-19 pandemic, leading to a subsequent contraction in their hiring. To assess such alternatives we consider various other robustness checks in Section 4.6, such as removing computer occupations and conditioning on whether the occupation is amenable to work from home.

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<sup>17</sup>Because of zero counts in the outcome variable, we estimate a Poisson regression instead of an OLS regression in logs following guidance from Chen and Roth (2024).

<sup>18</sup>Results are not sensitive to these restrictions, though there must be at least one non-zero value in each firm-month and each firm-quintile for observations to not get dropped in the Poisson regression.